

# Homework 3: RAG for Science Question Answering

RNN and Transformer Course

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## Abstract

This report presents the official Homework 3 pipeline using a 50-question random sample from the specified Kaggle `train.csv` file with `random_state=42`. The system compares two chunking strategies, builds two FAISS indices with BAAI/bge-m3, applies two-stage retrieval with dense search followed by `cross-encoder/ms-marco-MiniLM-L-6-v2`, and generates final answers with local Ollama llama3. On the official evaluation set, the final generation pipeline reached **80.00% accuracy (40/50)**. A previous science-only / DeepSeek / ensemble branch is retained only as an optional appendix and is not used as the formal HW3 score.

## Contents

|          |                                                     |           |
|----------|-----------------------------------------------------|-----------|
| <b>1</b> | <b>Introduction</b>                                 | <b>3</b>  |
| <b>2</b> | <b>Dataset and Official Evaluation Setup</b>        | <b>3</b>  |
| 2.1      | Random 50-question sample from train.csv . . . . .  | 3         |
| 2.2      | Wikipedia corpus construction . . . . .             | 3         |
| <b>3</b> | <b>Part 1: Indexing Pipeline</b>                    | <b>3</b>  |
| 3.1      | Chunking Strategy A . . . . .                       | 3         |
| 3.2      | Chunking Strategy B . . . . .                       | 4         |
| 3.3      | Vector Index Construction . . . . .                 | 4         |
| 3.4      | Chunk Size Analysis . . . . .                       | 4         |
| <b>4</b> | <b>Part 2: Retrieval and Re-ranking</b>             | <b>5</b>  |
| 4.1      | Dense Retrieval . . . . .                           | 5         |
| 4.2      | Cross-Encoder Re-ranking . . . . .                  | 5         |
| 4.3      | Hit Rate Comparison . . . . .                       | 5         |
| 4.4      | Query Formulation and Top-k Sensitivity . . . . .   | 6         |
| 4.5      | Two Qualitative Re-ranking Examples . . . . .       | 7         |
| <b>5</b> | <b>Part 3: Ollama Generation</b>                    | <b>9</b>  |
| 5.1      | Llama-3 Integration . . . . .                       | 9         |
| 5.2      | Anti-hallucination Prompt . . . . .                 | 9         |
| 5.3      | Accuracy on 50 Random train.csv Questions . . . . . | 9         |
| 5.4      | End-to-end Pipeline Ablation . . . . .              | 10        |
| <b>6</b> | <b>Latency Analysis</b>                             | <b>11</b> |

|          |                                                  |           |
|----------|--------------------------------------------------|-----------|
| <b>7</b> | <b>Error Analysis</b>                            | <b>11</b> |
| <b>8</b> | <b>Optional Advanced Experiment</b>              | <b>12</b> |
| 8.1      | Motivation . . . . .                             | 12        |
| 8.2      | Science-only evaluation setup . . . . .          | 12        |
| 8.3      | Advanced retrieval changes . . . . .             | 12        |
| 8.4      | Model and ensemble comparison . . . . .          | 13        |
| 8.5      | Why this is not the official HW3 score . . . . . | 13        |
| <b>9</b> | <b>Conclusion</b>                                | <b>14</b> |

# 1 Introduction

The goal of Homework 3 is to build a retrieval-augmented generation pipeline for science question answering. The official submission in this report follows the required structure:

- Part 1: compare two chunking strategies and build two vector indices.
- Part 2: implement dense retrieval plus cross-encoder re-ranking and compare hit rate.
- Part 3: use Ollama Llama-3 to answer multiple-choice questions from retrieved evidence.
- Latency analysis: measure vector search, re-ranking, and generation time on the same official pipeline.

## 2 Dataset and Official Evaluation Setup

### 2.1 Random 50-question sample from train.csv

The official evaluation set is created directly from the specified Kaggle train file:

```
train\_path = r"D:\\course\\rnnlstm\\HW3\\kaggle-llm-science-exam\\train.csv"
train\_df = pd.read\_csv(train\_path)
official\_eval\_df = train\_df.sample(n=50, random\_state=42).reset\_index(drop=True)
official\_eval\_df = official\_eval\_df.rename(columns={"prompt": "question"})
```

This produces a fixed 50-question sample from the 200 training rows in the specified file. This is the only dataset used for the formal HW3 results in this report.

### 2.2 Wikipedia corpus construction

The cached `official_wiki_articles.json` is used as the local Wikipedia plain-text corpus for this assignment run. Articles are converted into document objects before chunking.

| Metric                                | Value     |
|---------------------------------------|-----------|
| specified <code>train.csv</code> rows | 200       |
| Official evaluation rows              | 50        |
| Raw Wikipedia documents               | 500       |
| Total raw corpus characters           | 2,252,107 |

## 3 Part 1: Indexing Pipeline

### 3.1 Chunking Strategy A

Strategy A uses fixed-size recursive chunking with `chunk_size=500` and `overlap=50`. This is the smaller-granularity baseline index.

## 3.2 Chunking Strategy B

Strategy B uses larger recursive chunks with `chunk_size=1000` and `overlap=200`. This keeps more local context inside each chunk.

## 3.3 Vector Index Construction

Both indices use the same embedding model and GPU execution:

- Embedding model: BAAI/bge-m3
- Vector store: FAISS
- Device: CUDA GPU

Index A stores Strategy A chunks, and Index B stores Strategy B chunks.

## 3.4 Chunk Size Analysis

| Metric                       | Strategy A | Strategy B |
|------------------------------|------------|------------|
| Chunk size                   | 500        | 1000       |
| Overlap                      | 50         | 200        |
| Number of chunks             | 6,865      | 3,204      |
| Average chunk length (chars) | 334.48     | 732.73     |
| Maximum chunk length (chars) | 500        | 1000       |

Strategy A creates many shorter chunks, while Strategy B produces fewer but longer chunks. In this rerun, smaller chunks more frequently cut sentence boundaries and split complete answer spans across chunk edges, while Strategy B preserved longer contiguous evidence and gave more reliable complete-answer capture. The larger Strategy B chunks reduce fragmentation of multi-sentence explanations, which is important for science questions where the answer often depends on a definition plus a consequence. The tradeoff is that longer chunks may contain more irrelevant surrounding text, so re-ranking becomes more important.

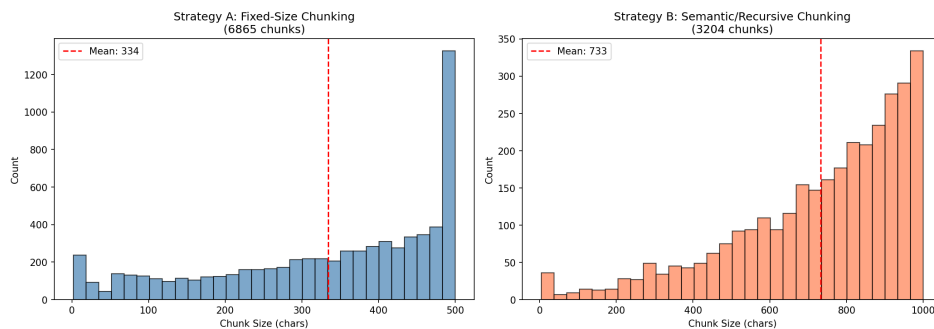


Figure 1: Chunk-size distribution from the notebook run.

## 4 Part 2: Retrieval and Re-ranking

### 4.1 Dense Retrieval

Dense retrieval is the high-recall first stage. It encodes the question into the same embedding space as the indexed chunks and uses vector similarity to quickly return top- $k$  candidates. This stage is computationally efficient and scales well, so it is suitable for broad candidate recall. The tradeoff is that embedding similarity is approximate: dense top-1 can be semantically adjacent or lexically similar, but still not the most answer-bearing chunk.

### 4.2 Cross-Encoder Re-ranking

Cross-encoder re-ranking is the high-precision second stage. For each query–chunk pair from the dense candidate set, the model jointly encodes both texts and scores relevance with token-level interactions. This usually gives finer discrimination than embedding-only retrieval, especially for ambiguous terms and dense top-1 ordering mistakes. It is slower than vector search, but the cost stays controlled because re-ranking is applied only to the dense top- $k$  candidates rather than the full index.

### 4.3 Hit Rate Comparison

Hit rate is reported as a retrieval proxy metric based on keyword overlap between the retrieved chunks and the correct answer text. The assignment requires this comparison, so it is kept explicitly as a proxy retrieval metric rather than a perfect relevance label. A question is counted as a hit if at least one token from the correct answer text (after stopword removal) appears in at least one of the final top-3 chunks. This is a proxy metric rather than a human relevance label.

| Index                        | Method                     | Hit Rate |
|------------------------------|----------------------------|----------|
| Index A (fixed 500/50)       | Vector Search Only         | 66.00%   |
| Index A (fixed 500/50)       | Vector Search + Re-ranking | 72.00%   |
| Index B (recursive 1000/200) | Vector Search Only         | 78.00%   |
| Index B (recursive 1000/200) | Vector Search + Re-ranking | 82.00%   |

This table is the main question-only retrieval comparison (`query_mode=question_only`, dense `top_k=50`, rerank `top_n=3`). Under this single consistent metric, re-ranking improves both indices (+6 pp on Index A and +4 pp on Index B), and Index B with re-ranking gives the highest proxy hit rate.

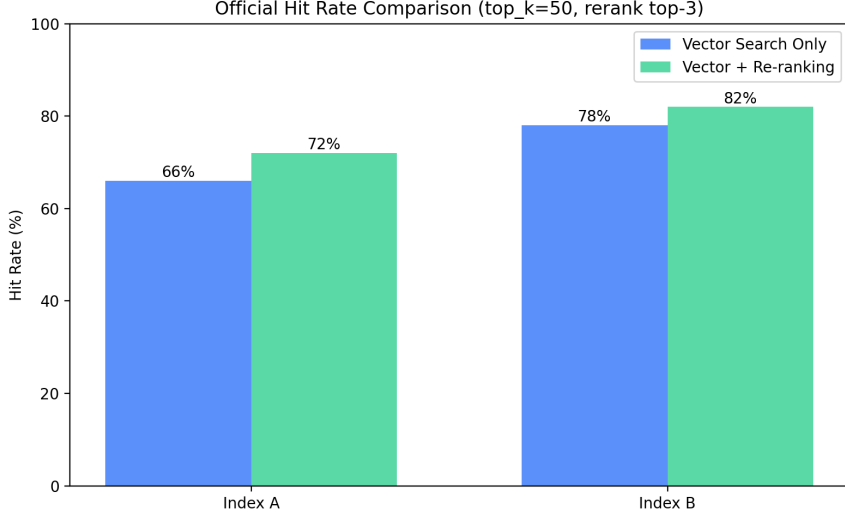


Figure 2: Official hit-rate comparison from the notebook run.

#### 4.4 Query Formulation and Top-k Sensitivity

We compared query construction and top-k settings on the same official random-50 split.

| Index | Query mode            | Vector-only | Reranked | Improvement |
|-------|-----------------------|-------------|----------|-------------|
| A     | question_only         | 66.00%      | 72.00%   | +6.00 pp    |
| A     | question_plus_choices | 78.00%      | 80.00%   | +2.00 pp    |
| B     | question_only         | 78.00%      | 82.00%   | +4.00 pp    |
| B     | question_plus_choices | 78.00%      | 84.00%   | +6.00 pp    |

On this split, `question_plus_choices` improves reranked hit rate, especially on Index B. However, this table is still a retrieval proxy; the official generation setting is selected by end-to-end QA accuracy rather than hit-rate proxy alone.

| top_k | Reranked hit rate | Accuracy | Rerank latency (s) | Total latency (s) |
|-------|-------------------|----------|--------------------|-------------------|
| 20    | 78.00%            | 70.00%   | 0.0153             | 2.3872            |
| 50    | 82.00%            | 74.00%   | 0.0334             | 2.4670            |

Increasing dense `top_k` from 20 to 50 improves reranked hit rate and final accuracy on the official split, while adding about 18 ms reranking time and about 80 ms total latency.

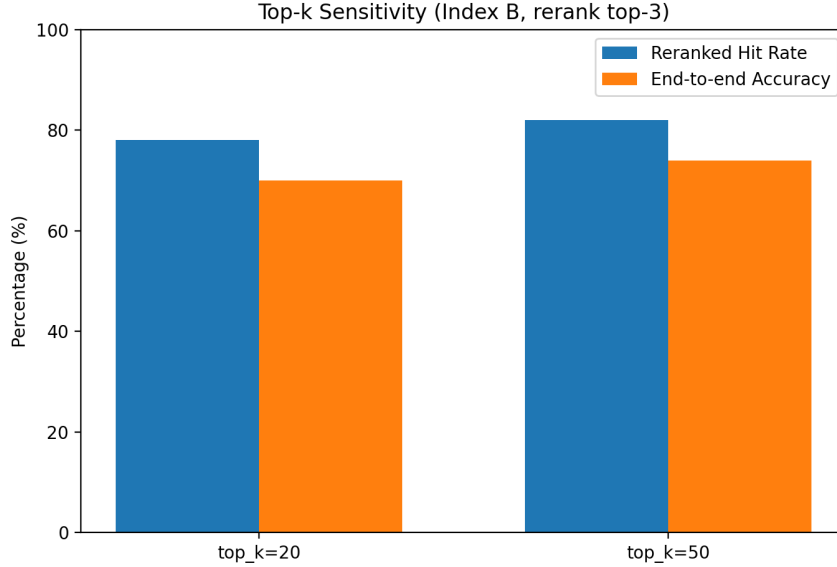


Figure 3: Top-k sensitivity on Index B with rerank top-3.

This top-k sensitivity table is an Index B retrieval ablation. The final official generation pipeline is selected separately by end-to-end QA accuracy (Section 5.4), which favors Index A + re-ranking in this run.

## 4.5 Two Qualitative Re-ranking Examples

The following two cases were selected from the official random-50 evaluation set using the final Index A retrieval branch. They show cases where vector search ranked a weak or irrelevant chunk first, while cross-encoder re-ranking promoted a more answer-relevant chunk to rank 1. These examples were selected by scanning the official random-50 set for cases where the dense top-1 chunk was off-topic and the cross-encoder promoted a higher-scoring, more answer-relevant chunk to rank 1.

### Case 1: Planetary-system concept promotion

- Source: official random-50
- Question: What is a planetary system?
- Correct answer: C. Any set of gravitationally bound non-stellar objects in or out of orbit around a star or star system.
- Vector Search top-1:
  - snippet: A planet is a large object such as Venus or Earth that orbits a star ...
  - score: 2.6344
  - why it is weak / irrelevant: this chunk defines a single planet and does not directly describe the broader planetary-system concept in the answer.
- Re-ranked top-1:

- snippet: Planetary astronomy is the study of planets ... and other small objects that orbit stars ...
- original vector rank: 2
- score: 4.2828
- evidence phrase: objects that orbit stars
- why it is more answer-relevant / supports the correct answer: it directly addresses the system-level definition around objects orbiting stars.
- Explanation: cross-encoder re-ranking promoted a broader, more answer-bearing chunk from rank 2 to rank 1 with a higher score.

## Case 2: Molecular-cloud formation mechanism promotion

- Source: official random-50
- Question: What is the reason for the formation of stars exclusively within molecular clouds?
- Correct answer: B. Low temperature and high density allow gravity to exceed outward pressure and trigger collapse.
- Vector Search top-1:
  - snippet: Many galaxies continue to form new generations of stars ...
  - score: -1.4520
  - why it is weak / irrelevant: this is broad star-formation background and misses the molecular-cloud collapse mechanism.
- Re-ranked top-1:
  - snippet: The Sun formed by gravity in a large molecular cloud ...
  - original vector rank: 25
  - score: -0.1704
  - evidence phrase: formed by gravity in a large molecular cloud
  - why it is more answer-relevant / supports the correct answer: it explicitly links star formation to molecular clouds and gravitational collapse context.
  - Explanation: cross-encoder re-ranking promoted a mechanism-bearing chunk from rank 25 to rank 1 with a clear score gain.

These examples are qualitative evidence-ordering demonstrations and do not change the official random-50 accuracy computation.



## 5 Part 3: Ollama Generation

### 5.1 Llama-3 Integration

The official generator is local Ollama llama3.

### 5.2 Anti-hallucination Prompt

The system prompt enforces two constraints:

- use only the retrieved context,
- output exactly one letter from A/B/C/D/E.

For open-ended RAG usage, the same policy maps insufficient evidence to “I do not know”, while this HW3 MCQ run enforces single-letter output.

The generator receives the reranked top-3 chunks together with the question and answer choices.

Prompt skeleton:

Context:

[retrieved chunks]

Question:

...

Choices:

A ...

B ...

C ...

D ...

E ...

Instruction:

Use only the context. Output exactly one letter.

### 5.3 Accuracy on 50 Random train.csv Questions

The official generation pipeline is:

- Index A
- dense retrieval top\_k=50
- cross-encoder re-ranking to top-3
- Ollama llama3

| Metric            | Value   |
|-------------------|---------|
| Official accuracy | 80.00%  |
| Correct answers   | 40 / 50 |

This is the formal HW3 score reported in the notebook and report.

The hit-rate experiment, accuracy evaluation, and latency analysis all use the same official retrieval configuration: `top_k=50` with cross-encoder re-ranking to top-3.

## 5.4 End-to-end Pipeline Ablation

| Pipeline                       | Retrieval setting             | Correct / 50 | Accuracy |
|--------------------------------|-------------------------------|--------------|----------|
| Index A + Vector-only + Llama3 | dense top-3                   | 37 / 50      | 74.00%   |
| Index A + Re-ranking + Llama3  | dense top-k=50 + rerank top-3 | 40 / 50      | 80.00%   |
| Index B + Vector-only + Llama3 | dense top-3                   | 35 / 50      | 70.00%   |
| Index B + Re-ranking + Llama3  | dense top-k=50 + rerank top-3 | 37 / 50      | 74.00%   |

Re-ranking improves end-to-end QA accuracy for both indices in this ablation. Although Index B gives the strongest retrieval proxy hit rate, the final pipeline is selected by end-to-end QA accuracy. Therefore, the official generation pipeline uses Index A + re-ranking. Because the hit-rate metric is a lightweight keyword-overlap proxy, it does not perfectly predict final multiple-choice accuracy.

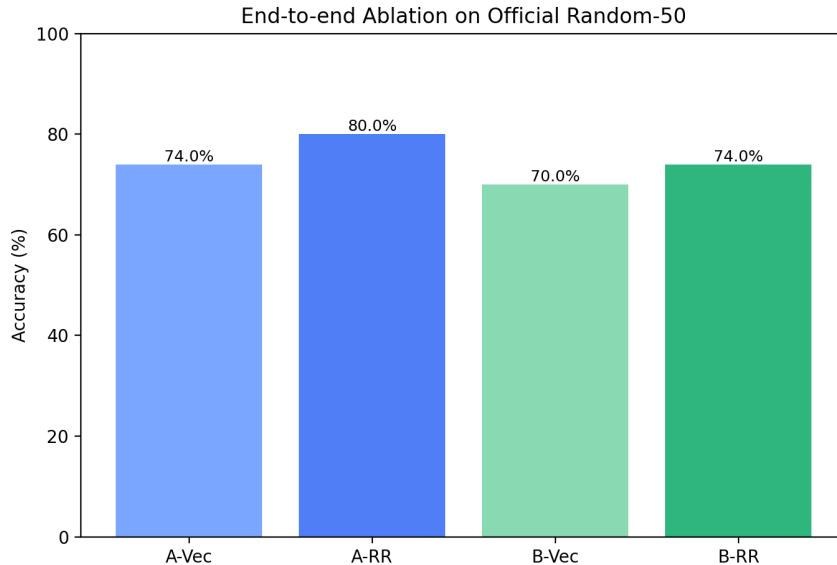


Figure 4: End-to-end pipeline ablation on the official random-50 split.

## 6 Latency Analysis

Latency is measured on the same official pipeline and evaluation set.

| Stage          | Average Latency (s) |
|----------------|---------------------|
| Vector search  | 0.0132              |
| Re-ranking     | 0.0269              |
| LLM generation | 2.4215              |
| Total          | 2.4616              |

Re-ranking adds about 26.9 milliseconds on average, which is still very small compared with the 2.422-second LLM generation stage. Re-ranking accounts for approximately **1.09%** of total latency ( $0.0269/2.4616$ ), so most runtime remains dominated by LLM generation.

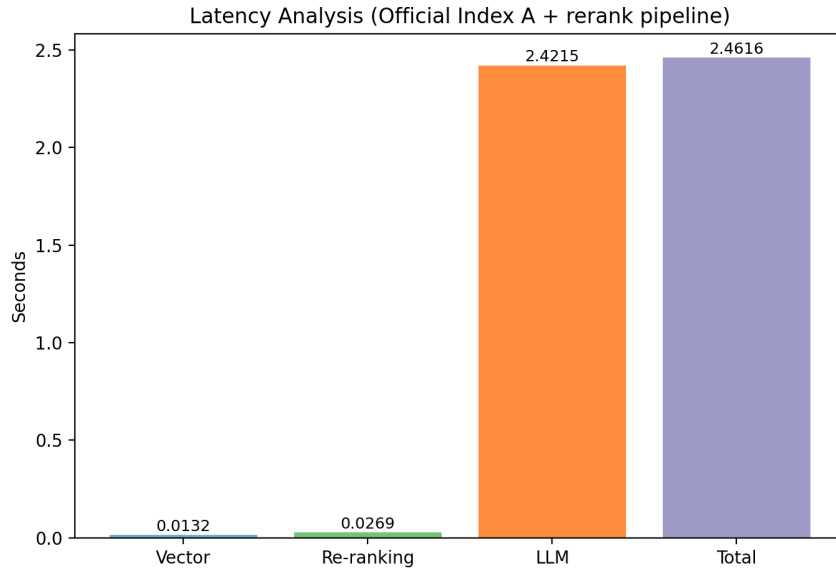


Figure 5: Latency breakdown from the official notebook run.

## 7 Error Analysis

The official pipeline (Index A + rerank + Llama3, top-k=50) produced 10 errors out of 50 questions. Error categories are summarized below.

| Error category          | Count |
|-------------------------|-------|
| Missing evidence        | 4     |
| Partial / weak evidence | 1     |
| LLM reasoning error     | 5     |

Representative pattern 1: retrieved chunks are on-topic but incomplete, so the model selects a plausible distractor. Representative pattern 2: retrieved top-3 contains useful

terms, but Llama3 still maps evidence to the wrong option. Likely improvements are stronger evidence coverage, choice-aware retrieval signals, and tighter answer-calibration prompts.

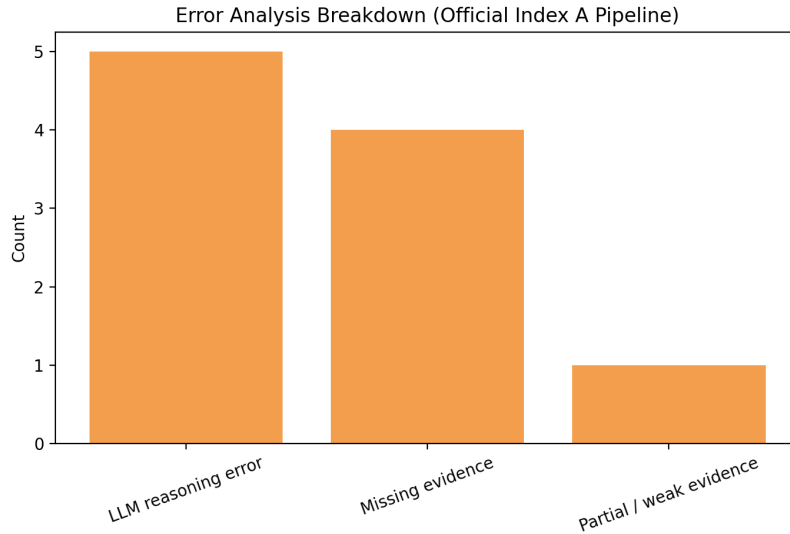


Figure 6: Official error analysis breakdown.

## 8 Optional Advanced Experiment

### 8.1 Motivation

The official random 50-question `train.csv` sample remains the required HW3 setting. The appendix branch was tested only to study whether stronger retrieval, answer-aware evidence scoring, and model variants improve science-only QA.

### 8.2 Science-only evaluation setup

The advanced branch filtered the public mirror with science keywords, creating a science-only pool with the following traceable setup from `advanced_experiment_summary.json`:

| Metric                     | Value |
|----------------------------|-------|
| Public mirror rows         | 6,684 |
| Science-only pool rows     | 1,469 |
| Science dev rows           | 10    |
| Science held-out test rows | 50    |

This appendix-only setup is not the official HW3 evaluation because it does not use the required random 50-question `train.csv` sample.

### 8.3 Advanced retrieval changes

The advanced notebook branch (`hw3advance.ipynb`) implemented:

- query rewriting for Wikipedia retrieval,
- long-form Wikipedia evidence extraction,
- chunk-level re-ranking over fetched evidence,
- option-aware evidence scoring,
- NLI-style scoring with a DeBERTa entailment model.

## 8.4 Model and ensemble comparison

The following advanced appendix numbers are traceable from `advanced_experiment_summary.json` and `advanced_rag_results.csv`:

| Appendix experiment                                  | Accuracy |
|------------------------------------------------------|----------|
| Llama 3 model-only on corrected science dev slice    | 30.00%   |
| DeepSeek model-only on corrected science dev slice   | 40.00%   |
| DeepSeek RAG vector-only                             | 30.00%   |
| DeepSeek RAG with re-ranking                         | 60.00%   |
| Option-aware rerank + NLI                            | 60.00%   |
| Final ensemble on 10-question dev slice              | 80.00%   |
| Final ensemble on held-out 50 science-only questions | 72.00%   |
| Post-hoc best rule on the same science-only run      | 74.00%   |

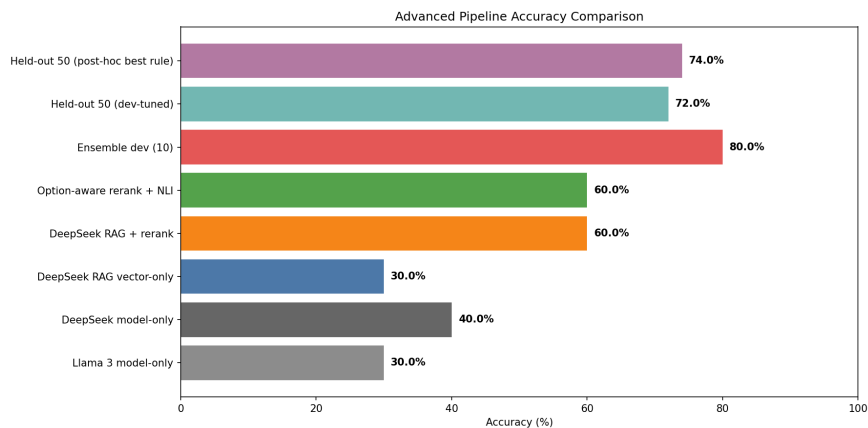


Figure 7: Appendix-only advanced method comparison from notebook outputs.

## 8.5 Why this is not the official HW3 score

These results are useful as an ablation and extension appendix, but they are not reported as the formal HW3 score because:

- they use a filtered science-only pool rather than the required random 50-question `train.csv` sample,

- they use DeepSeek and ensemble extensions beyond the official Ollama llama3 pipeline,
- the formal HW3 result in this report remains the official **80.00%** score on the `random-train.csv` evaluation.

## 9 Conclusion

The final official HW3 pipeline uses a random 50-question sample from the specified Kaggle `train.csv` path, two chunking strategies, FAISS retrieval, cross-encoder re-ranking, and Ollama llama3 generation on GPU. The final official accuracy is **80.00% (40/50)**. Strategy B gives the strongest retrieval proxy hit rate, but the end-to-end ablation shows the best final QA accuracy comes from Index A with re-ranking, so that setting is selected for the formal result. The appendix shows that a science-only DeepSeek/ensemble branch can reach a higher ceiling, but the formal HW3 score reported here remains 80.00%.